

Bruno Borges de Souza

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Summary

Data scientist, graduated in production engineering and applied mathematics.

AWS Machine Learning Specialty Certification.

Experience with supervised and unsupervised machine learning applications for creating recommender, classification and clustering systems.

Experience with python and R languages, using several libraries related to data science (Pandas, numpy, scikit-learn, scipy, pyspark, tensorflow, lightFM etc).

Advanced knowledge in statistics (multivariate analysis, descriptive statistics, inference and hypothesis testing).

Experience



Cogitum Analytica

Dez 2023 – Current (Freelancer)

From initial data exploration to building and assessing predictive models using statistical techniques and machine learning algorithms.

Delivering tailored solutions addressing specific business challenges for each client.



Data Scientist

DigitalReef

Jul 2022 - Jul 2023 (1 year 9 months)

- Develop Machine Learning models and apply them to business problems;
- Collect, clean and organize data from different sources;
- Maintenance of the campaign optimization system;
- Improvement of machine learning models using feature embedding techniques, bayesian optimization for tuning hyperparameters and deep learning;
- Implementation of a data lake using AWS Lake Formation, Glue and open source Data Lakehouse tools (Iceberg, Hudi and delta lake);
- Creation of recommender systems using collaborative filtering.



Data Scientist

Banco Pan

Nov 2021 – Jul 2022 (9 months)

- Conducted detailed analyses of diverse datasets such as demographic information, credit history, payment behavior, and macroeconomic data.
- Identified weaknesses in the credit portfolio, detected market trends, and pinpointed business opportunities.
- Actively participated in the deployment process of predictive services, adhering to MLOps best practices. This involved automating data pipelines, model selection and training, and continuous performance monitoring using techniques like cross-validation and metrics such as AUC-ROC and Gini coefficient.
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Data Scientist

Lumini IT Solutions

Apr 2020 - Nov 2021 (1 year 8 months)

- Development of statistical and mathematical analysis, optimization and other problems involving data and analytics;
- Detect unusual patterns and business-relevant data;
- Create machine learning models to classify, group and recommend leads;
- Create dashboards in Power Bi and present information using data visualization techniques (Plotly, seaborn, matplotlib);
- Implementation of machine learning models in the AWS environment;

The lead recommendation system brought real results for the client, with conversions increasing by up to 50% in some sales channels.



Fellow Masters

CAPES - Coordenação de Aperfeiçoamento de Pessoal de Nível Superior

Sep 2017 - Aug 2019 (2 years)

- Research equilibrium point properties in fractional ordinary differential equations, particularly in applications to fractional neural networks";
- Consultancy in data science projects focused on Monte Carlo simulation, investment portfolio optimization and creation of machine learning and deep learning models for a credit score modeling area.

Education



Instituto de Matemática e Estatística - Universidade de São Paulo (IME-USP)

Master's Degree, Applied Mathematics

2017 - 2019

Dissertation: Stability of Hopfield Neural Networks with Derivative in Fractional Order. Theory of Control, Optimization, Machine Learning to Finance, Differential Equations, Real Analysis, Linear Algebra, Advanced Calculus, Artificial Neural Networks, Differential Evolution.



Anhembi Morumbi University

Bachelor's degree, Production Engineering

2012 - 2016

- Monograph: Application of Markowitz Portfolio Theory to optimization of a fixed and variable income portfolio;
- ProUni scholarship holder;
- Voluntary Monitor in the discipline of Differential Equations Ordinary.



USP - Universidade de São Paulo

Scientific Initiation, Applied Mathematic

Licenses & Certifications



Desenvolvimento de Habilidades Orais e Pronúncia da Língua Inglesa nível B2 -
Ministério da Educação
8455FCDE



Machine Learning - Itaú Unibanco



Microsoft Power BI para Data Science - Data Science Academy
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AWS Certified Cloud Practitioner - Amazon Web Services (AWS)
Issued Aug 2022 - Expires Aug 2025



AWS Certified Machine Learning – Specialty - Amazon Web Services (AWS)
Issued Oct 2022 - Expires Oct 2025

Skills

Data Lakes • MLOps • Amazon Web Services • Python • Aprendizado de máquina • Machine Learning • Mathematics Education • Python (Programming Language) • Gestão de projetos • Engenharia

Honors & Awards



National Bronze Medal - Brazilian Mathematical Olympiad - IMPA

Jun 2011

Concerning the performance, where I obtained bronze medal, national, in OBMEP. The Brazilian Public Mathematics Olympiad (OBMEP) is an achievement of the National Institute of Pure and Applied Mathematics - IMPA - and aims to stimulate the study of mathematics and reveal talents in the area.



Regional Gold in the Brazilian Mathematical Olympiad

Jan 2011